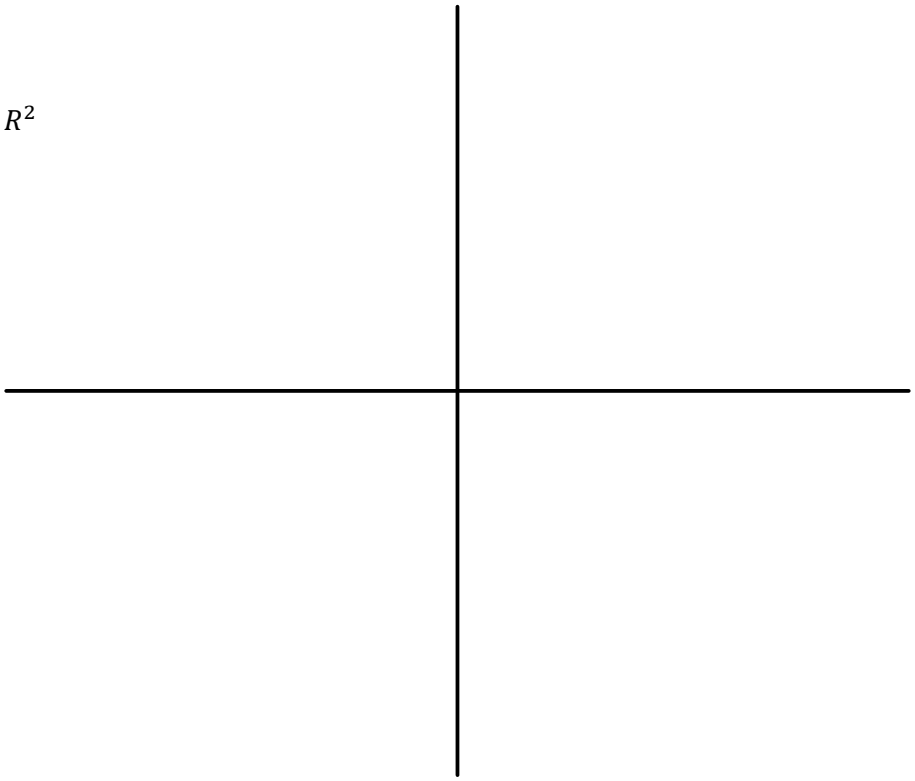
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Before we talk about inner product spaces it is useful to review the idea of a "basis" , "orthogonal basis" and "orthogonal projection".

In the context of vectors a basis can be defined as a set of linearly independent vectors which span the space that they belong to.

Ex. The vectors <-1,1> and <0,1> form a basis for R^2



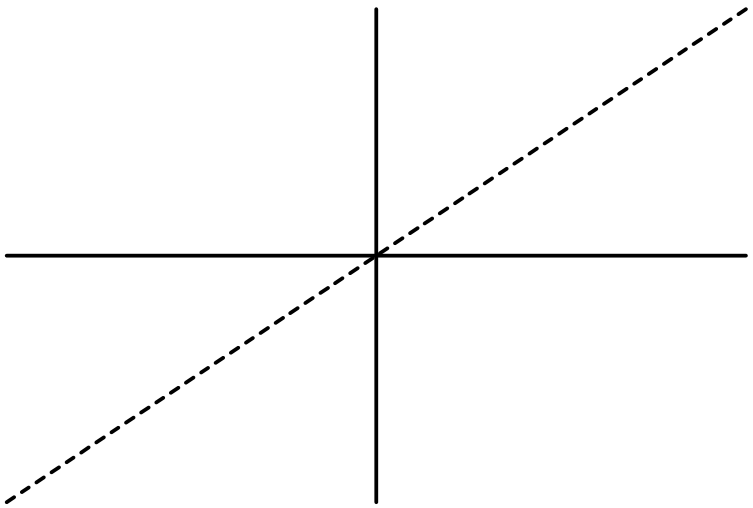
Linear Independence:

They "span" R^2

An orthogonal basis is a basis of vectors for a given space which are orthogonal.

Example: <1,0,0> , <0,1,0> ,<0,0,1> form an orthogonal basis for R^3

Let's show that they are orthogonal.



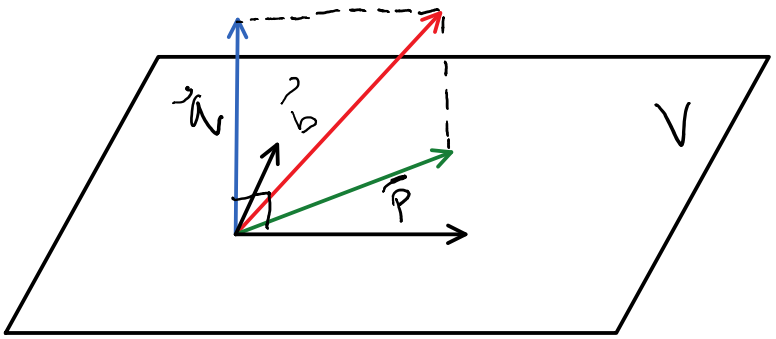
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Orthogonal Projection Problem:

Given a vector $\vec{b}$ not in a subspace V, find vectors $\vec{p}$ in V and $\vec{q}$ orthognal to V such that $\vec{b} = \vec{p} + \vec{q}$.

Further we wish to represent $\vec{p}$ with the basis of V $\{\overrightarrow{a_0}, \overrightarrow{a_1}, \overrightarrow{a_2}, \cdots, \overrightarrow{a_n}\}$ *idealy an orthogonal basis.*



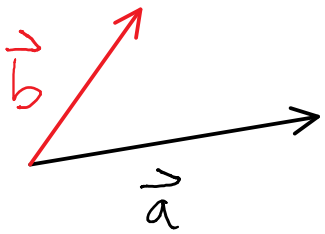
Notice that $\vec{p}$ is the closest vector to $\vec{b}$ that lies in V

Let's first suppose that we have an orthogonal basis for a subspace V $\{\overrightarrow{a_0}, \overrightarrow{a_1}, \overrightarrow{a_2}, \cdots, \overrightarrow{a_n}\}$ This means that $\overrightarrow{a_i} \cdot \overrightarrow{a_j} = 0$ if $i \neq j$

Recall that the projection of one vector onto another is defined by $comp_{\vec{a}}(\vec{b}) = \frac{\vec{a}\cdot\vec{b}}{||\vec{a}||}$

.

That is it is a measure of how much of the vector $\vec{b}$ is in the direction of $\vec{a}$

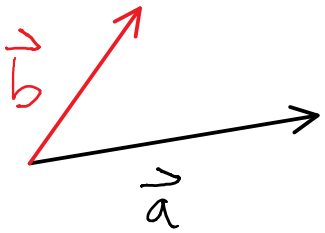


We could then take this amount and scale $\vec{a}$ to make a vector in the same direction as $\vec{a}$ call it $\vec{p}$

How? Well take $\vec{a}$ and divide it by its magnitude to create a unit vector in the same direction $\frac{\vec{a}}{||\vec{a}||}$

then multiply it by $\vec{b}$ is in the directioon of $\vec{a}$ namely the scalar we get from $comp_{\vec{a}}(\vec{b}) = \frac{\vec{a}\cdot\vec{b}}{||\vec{a}||}$

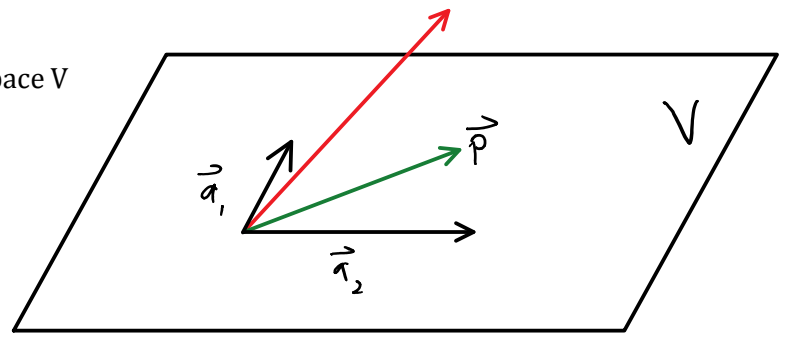
We then have $$\vec{p} = \frac{\vec{a}\cdot\vec{b}}{||\vec{a}||}\,\frac{\vec{a}}{||\vec{a}||} = \frac{\vec{a}\cdot\vec{b}}{\vec{a}\cdot\vec{a}}\,\vec{a}$$



It turns out that if we do this to $\vec{b}$ with each vector in our orthogonal basis $\{\overrightarrow{a_0}, \overrightarrow{a_1}, \overrightarrow{a_2}, \cdots, \overrightarrow{a_n}\}$ and add them up, we end up with the orghognal projection of $\vec{b}$ on to the subspace V interms of the orthogonal basis of V.

The orthogonal projection of the vector $\vec{b}$ on to the subspace V Is given by:

$$\vec{p} = \frac{\overrightarrow{a_1}\cdot\vec{b}}{\overrightarrow{a_1}\cdot\overrightarrow{a_1}}\overrightarrow{a_1} + \frac{\overrightarrow{a_2}\cdot\vec{b}}{\overrightarrow{a_2}\cdot\overrightarrow{a_2}}\overrightarrow{a_2} + \cdots + \frac{\overrightarrow{a_n}\cdot\vec{b}}{\overrightarrow{a_n}\cdot\overrightarrow{a_n}}\overrightarrow{a_n}$$



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It turns out that the vector $\vec{p}$ that we found is the best approximation of $\vec{p}$ using the basis of V $\{\overrightarrow{a_0}, \overrightarrow{a_1}, \overrightarrow{a_2}, \cdots, \overrightarrow{a_n}\}$

While it may not seem particularly useful to approximate a known vector it does have its uses in inverse problems where we can't find an exact answer. For now though we might consider the case where $\vec{b}$ lies in the subspace V. In this case $\vec{p} = \vec{b}$ and our formula decomposes $\vec{b}$ into a linear combo of our orthognal basis. This idea becomes most useful for PDE's when we extend our notion of a "vector space" and a "dot product".

Definition: A *vector space* is a set V on which two operations + and · are defined, called *vector addition* and *scalar multiplication.* **Note this · is not the dot product here**

The operation + (vector addition) must satisfy the following conditions:

Closure: If **u** and **v** are any vectors in V, then the sum **u** + **v** belongs to V.

(1) *Commutative law:* For all vectors **u** and **v** in V, **u** + **v** = **v** + **u**

(2) *Associative law:* For all vectors **u**, **v**, **w** in V, **u** + (**v** + **w**) = (**u** + **v**) + **w**

(3) *Additive identity:* The set V contains an *additive identity* element, denoted by **0**, such that for any vector **v** in V, **0** + **v** = **v** and **v** + **0** = **v**.

(4) *Additive inverses:* For each vector **v** in V, the equations **v** + **x** = **0** and **x** + **v** = **0** have a solution **x** in V, called an *additive inverse* of **v**, and denoted by - **v**.

The operation · (scalar multiplication) is defined between real numbers (or scalars) and vectors, and must satisfy the following conditions:

Closure: If **v** in any vector in V, and c is any real number, then the product c · **v** belongs to V.

(5) *Distributive law:* For all real numbers c and all vectors **u**, **v** in V, c · (**u** + **v**) = c · **u** + c · **v**

(6) *Distributive law*: For all real numbers c, d and all vectors **v** in V, (c+d) · **v** = c · **v** + d · **v**

(7) *Associative law*: For all real numbers c,d and all vectors **v** in V, c · (d · **v**) = (cd) · **v**

(8) *Unitary law*: For all vectors **v** in V, 1 · **v** = **v**

Notice in the above definition V was just a set of general objects not necessarily vectors as we have used them.

In fact we can actually define a vector space of functions. For example let's take the set of all polynomials P on the real line and define a vector space. Let $p(x), q(x) \in P$

Vector addition: $(p+q)(x) = p(x) + q(x)$ Scalar multiplication: $c \cdot f(x) = cf(x)$

Let's just check a few properties:

Is it closed under addition and scalar multiplication?

(1)

(2)

(3)

(4)

(5)

(6)

(7)

(8)

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What about a basis for P? Well recall that a basis is a set of objects in our set which are linearly independent and span the set. For the set of all polynomials one basis is:

$$\{1, x, x^2, x^3, x^4, \cdots\}$$

Are they linearly independent?

Do they span the set of all polynomials?

Ok so we have a basis for our vector space of functions. One natural question now is what does our "dot product" look like?

Well it turns out that the dot product is a special case of what we call an "inner product"

An **inner product** on a vector space V is a function that associates with each pair of vectors **u** and **v** in V a scalar number <**u** ,**v**> such that:

(1) $< \boldsymbol{u}, \boldsymbol{v} > = < \boldsymbol{v}, \boldsymbol{u} >$

(2) $< \boldsymbol{u}, \boldsymbol{v} + \boldsymbol{w} > = < \boldsymbol{u}, \boldsymbol{v} > + < \boldsymbol{u}, \boldsymbol{w} >$

(3) $< \boldsymbol{cu}, \boldsymbol{v} > = \boldsymbol{c} < \boldsymbol{u}, \boldsymbol{v} >$

(4) $< \boldsymbol{u}, \boldsymbol{u} > \geq 0 \; and < \boldsymbol{u}, \boldsymbol{u} > = 0 \;\; if \; and \; only \; if \; \boldsymbol{u} = \boldsymbol{0}$

If we have a vector space V with together with an inner product we call this an inner product space. For function vector spaces our inner products are usually integrals of products of functions.

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An example of an inner product space: Let $P^n[-1,1]$ be the set of continuious polynomials of degree at most n on the interval $[-1,1]$.

We define the inner product between two polynomials $p(x), q(x) \in P^n[-1,1]$ to be:

$$< p(x), q(x) > = \int_{-1}^{1} p(x)q(x)dx$$

Let's verify that this is indeed an inner product:

(1) $< \boldsymbol{u}, \boldsymbol{v} > = < \boldsymbol{v}, \boldsymbol{u} >$

(2) $< \boldsymbol{u}, \boldsymbol{v} + \boldsymbol{w} > = < \boldsymbol{u}, \boldsymbol{v} > + < \boldsymbol{u}, \boldsymbol{w} >$

(3) $< \boldsymbol{cu}, \boldsymbol{v} > = \boldsymbol{c} < \boldsymbol{u}, \boldsymbol{v} >$

(4) $< \boldsymbol{u}, \boldsymbol{u} > \geq 0 \; and < \boldsymbol{u}, \boldsymbol{u} > = 0 \;\; if \; and \; only \; if \; \boldsymbol{u} = \boldsymbol{0}$

In checking (4) we might think we can use the inner product to measure the "length" of a function in fact we can, we define the norm of a function f on our space as:

$$||f|| = \left(\int_{-1}^{1} \left(f(x) \right)^2 \right)^{\frac{1}{2}} = \sqrt{< f, f >}$$

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As we stated before the dot product we are used to is simply a special case of an **inner product** we might also recall that for conventional vectors if their dot product is zero they are orthogonal.

We extend this notion to all inner product spaces and say that:

Two general vectors $\boldsymbol{u}\ and\ \boldsymbol{v}$ are orthognal if and only if $< \boldsymbol{u}, \boldsymbol{v} > = \boldsymbol{0}$

Let's see if our basis for the set of all polynomials on $[-1,1]$ $\{1, x, x^2, x^3, x^4, \cdots\}$ is an orthogonal basis:

Let's try x^2 and x

Let's try $x^2$and x^4

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For reasons that will become apparent in a bit we prefer to have orthogonal bases. Thanks to two math-magicians, Gram and Schmidt, we have an algorithm for turning any given basis into an orthogonal basis.

Gram-Schmidt Orthogonalization:

Given any set of linearly independent vectors $v_1, v_2, v_3 \cdots v_n$ we can obtain a set of orthogonal vectors $u_1, u_2, u_3, \cdots u_n$ that span the same space with the following algorighim.

$u_1 = v_1$

$$u_2 = v_2 - \frac{< u_1, v_2 >}{< u_1, u_1 >} u_1$$

$$u_3 = v_3 - \frac{< u_1, v_2 >}{< u_1, u_1 >} u_1 - \frac{< u_2, v_3 >}{< u_2, u_2 >} u_2$$

$\vdots$

$\vdots$

$$u_n = v_n - \frac{< u_1, v_2 >}{< u_1, u_1 >} u_1 - \frac{< u_2, v_3 >}{< u_2, u_2 >} u_2 - \cdots - \frac{< u_{n-1}, v_n >}{< u_{n-1}, u_{n-1} >} u_{n-1}$$

In order to get a sense for why this works let's think of our conventional vectors. For the vectors we are used to the inner product becomes a dot product.

$u_1 = v_1$

$$u_2 = v_2 - \frac{u_1 \cdot v_2}{u_1 \cdot u_1} u_1$$

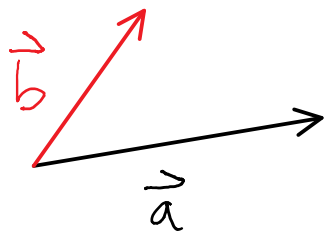
$$u_3 = v_3 - \frac{u_1 \cdot v_2}{u_1 \cdot u_1} u_1 - \frac{u_2 \cdot v_3}{u_2 \cdot u_2} u_2$$

$\vdots$

$\vdots$

$$u_n = v_n - \frac{u_1 \cdot v_2}{u_1 \cdot u_1} u_1 - \frac{u_2 \cdot v_3}{u_2 \cdot u_2} u_2 - \cdots - \frac{u_{n-1} \cdot v_n}{u_{n-1} \cdot u_{n-1}} u_{n-1}$$

$$\vec{p} = \frac{\vec{a} \cdot \vec{b}}{||\vec{a}||} \frac{\vec{a}}{||\vec{a}||} = \frac{\vec{a} \cdot \vec{b}}{\vec{a} \cdot \vec{a}} \vec{a}$$



So for each new basis vector u_i we are just removing the bits that are parallel to v_i ! It is much the same with the general algothrim we are removing the bits that are not orthognal.

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If we apply this algorithm to our set of basis vectors for $P^n[-1,1]$ $\{1, x, x^2, x^3, x^4, \cdots\}$

we obtain multiples of a famous set of polynomials called Legendre Polynomials of which the first 6 are:

$p_0 = 1$

$p_1 = x$

$p_2 = \frac{1}{2}(3x^2 - 1)$

$p_3 = \frac{1}{2}(5x^3 - 3x)$

$p_4 = \frac{1}{8}(35x^4 - 30x^2 + 3)$

$p_5 = \frac{1}{8}(63x^5 - 70x^3 + 15x)$

Gram-Schmidt Orthogonalization:

Given any set of linearly independent vectors $v_1, v_2, v_3 \cdots v_n$ we can obtain a set of orthogonal vectors $u_1, u_2, u_3, \cdots u_n$ that span the same space with the following algorighim.

$u_1 = v_1$

$u_2 = v_2 - \frac{< u_1, v_2 >}{< u_1, u_1 >} u_1$

$u_3 = v_3 - \frac{< u_1, v_2 >}{< u_1, u_1 >} u_1 - \frac{< u_2, v_3 >}{< u_2, u_2 >} u_2$

$\vdots$

$\vdots$

$u_n = v_n - \frac{< u_1, v_2 >}{< u_1, u_1 >} u_1 - \frac{< u_2, v_3 >}{< u_2, u_2 >} u_2 - \cdots - \frac{< u_{n-1}, v_n >}{< u_{n-1}, u_{n-1} >} u_{n-1}$

Let's calculate the first two, you get to do the rest in your homework. :)

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Ok so what is the point of all of this math... Recall earlier that if we had a vector $\vec{p}$ in a space V we could decompose it into a linear combo the orthognal basis vectors for V with the formula

$$\vec{p} = \frac{\overrightarrow{a_1} \cdot \vec{b}}{\overrightarrow{a_1} \cdot \overrightarrow{a_1}} \overrightarrow{a_1} + \frac{\overrightarrow{a_2} \cdot \vec{b}}{\overrightarrow{a_2} \cdot \overrightarrow{a_2}} \overrightarrow{a_2} + \cdots + \frac{\overrightarrow{a_n} \cdot \vec{b}}{\overrightarrow{a_n} \cdot \overrightarrow{a_n}} \overrightarrow{a_n}$$

It turns out we can do the same thing with functions! All we have to do is replace the dot product with the appropriate inner product that is for a given function $f(x) \in C([a,b])$ (the set of continuious functions on the interval $[a,b]$) if we have an orthognal basis for $C([a,b])$ $\{g_1(x), g_2(x), \cdots\}$

We can write:

$$f(x) = \frac{< g_1, f >}{< g_1, g_1 >} g_1 + \frac{< g_2, f >}{< g_2, g_2 >} g_2 + \cdots + \frac{< g_n, f >}{< g_n, g_n >} g_n + \cdots$$

This means that we can represent a given function as a linear combination of basis functions!

We have actually seen this already, if you recall we were able to represent our initial conditions to the heat equation as:

For homogeneous Dirichlet boundary conditions we had

$$u(x,0) = f(x) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi}{L}x\right)$$

Where

$$B_n = \frac{2}{L}\int_0^L f(x) \sin\left(\frac{n\pi}{L}x\right) dx = \frac{\int_0^L f(x) \sin\left(\frac{n\pi}{L}x\right) dx}{\int_0^L \sin^2\left(\frac{n\pi}{L}x\right) dx} = \frac{< \sin\left(\frac{n\pi x}{L}\right), f(x) >}{< \sin\left(\frac{n\pi x}{L}\right), \sin\left(\frac{n\pi x}{L}\right) >}$$

If we define the inner product

$< f, g > = \int_0^L f(x)g(x)dx$ we are simply saying: $f(x) = \frac{< g_1, f >}{< g_1, g_1 >} g_1 + \frac{< g_2, f >}{< g_2, g_2 >} g_2 + \cdots + \frac{< g_n, f >}{< g_n, g_n >} g_n + \cdots$

With $g_n = \sin\left(\frac{n\pi x}{L}\right)$ which are indeed orthognal since we we showed that:

$$< \sin\left(\frac{n\pi x}{L}\right), \sin\left(\frac{m\pi x}{L}\right) > = \int_0^L \sin\left(\frac{n\pi}{L}x\right) \sin\left(\frac{m\pi}{L}x\right) dx = \begin{cases} 0 & when\ m \neq n \\ \frac{L}{2} & when\ m = n \end{cases}$$

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It turns out that an orthogonal basis for the set of piecewise smooth functions on the interval $[-L, L]$ is:

$$\{1, \sin\left(\frac{\pi x}{L}\right), \sin\left(\frac{2\pi x}{L}\right), \cdots, \cos\left(\frac{\pi x}{L}\right), \cos\left(\frac{2\pi x}{L}\right), \cdots\}$$

This means if we have any piece wise smooth function on the interval [-L,L] we could represent it as a sum of a constant and sines and cosines with the appropriate coefficients determined by an inner product.

For the interval [-L,L] we would use the innerproduct $< f, g > = \int_{-L}^{L} f(x)g(x)dx$

That is we can write

$$f(x)$$
$$= \frac{< 1, f >}{< 1,1 >} 1 + \frac{< \sin\left(\frac{\pi x}{L}\right), f >}{< \sin\left(\frac{\pi x}{L}\right), \sin(\frac{\pi x}{L}) >} \sin\left(\frac{\pi x}{L}\right) + \frac{< \sin\left(\frac{2\pi x}{L}\right), f >}{< \sin\left(\frac{2\pi x}{L}\right), \sin(\frac{2\pi x}{L}) >} \sin\left(\frac{2\pi x}{L}\right) + \cdots$$
$$+ \frac{< \cos\left(\frac{\pi x}{L}\right), f >}{< \cos\left(\frac{\pi x}{L}\right), \cos\left(\frac{\pi x}{L}\right) >} \cos\left(\frac{\pi x}{L}\right) + \frac{< \cos\left(\frac{2\pi x}{L}\right), f >}{< \cos\left(\frac{2\pi x}{L}\right), \cos\left(\frac{2\pi x}{L}\right) >} \cos\left(\frac{2\pi x}{L}\right) + \cdots$$

And

$$\frac{<1, f>}{<1,1>} = \frac{\int_{-L}^{L} f(x)dx}{\int_{-L}^{L} dx} = \frac{\int_{-L}^{L} f(x)}{2L} = \frac{1}{L}\int_{-L}^{L} f(x)dx = A_0$$

$$\frac{< \sin\left(\frac{n\pi x}{L}\right), f >}{< \sin\left(\frac{n\pi x}{L}\right), \sin\left(\frac{n\pi x}{L}\right) >} = \frac{\int_{-L}^{L} f(x) \sin\left(\frac{n\pi x}{L}\right) dx}{\int_{-L}^{L} \sin^2\left(\frac{n\pi x}{L}\right) dx} = \frac{\int_{-L}^{L} f(x) \sin\left(\frac{n\pi x}{L}\right) dx}{L} = \frac{1}{L}\int_{-L}^{L} f(x) \sin\left(\frac{n\pi x}{L}\right) dx = B_n$$

$$\frac{< \cos\left(\frac{n\pi x}{L}\right), f >}{< \cos\left(\frac{n\pi x}{L}\right), \cos\left(\frac{n\pi x}{L}\right) >} = \frac{\int_{-L}^{L} f(x) \cos\left(\frac{n\pi x}{L}\right) dx}{\int_{-L}^{L} \cos^2\left(\frac{n\pi x}{L}\right) dx} = \frac{\int_{-L}^{L} f(x) \cos\left(\frac{n\pi x}{L}\right) dx}{L} = \frac{1}{L}\int_{-L}^{L} f(x) \cos\left(\frac{n\pi x}{L}\right) dx = A_n$$

So we can then represent our sum in the more familiar form we saw with periodic boundary conditions of

$$f(x) = A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi}{L}x\right) + \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi}{L}x\right)$$

With this choice of basis functions we call this a Fourier series!

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